

26 Platinum is one of the catalysts used in a catalytic converter.

Which statements about the platinum catalyst are correct?

- 1 More sites are available for catalysis to occur in 1.0 g of spherical nanoparticles with a diameter of 5 nm compared to 1.0 g of spherical nanoparticles with a diameter of 10 nm.
- 2 Platinum nanoparticles react with nitrogen oxide to form nitrogen and platinum oxide.
- 3 Nanoparticles of platinum are used since they have a large surface area to volume ratio.

A 1, 2 and 3

B 1 and 3 only

C 2 only

D 3 only